

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	SWTID-2026-8260
Project Name	A Comprehensive Measure of Well-Being using HDI Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Proper indentation used throughout Python code	Yes	Clean structured code
2	Proper File Structure	Separate files for model, preprocessing, and web app	Yes	Separate files for model, app
3	Meaningful Variable Names	Variables named based on HDI features and data	Yes	Easy to understand
4	Function / Method Names	Functions clearly represent their purpose (train, predict)	Yes	Clearly defined
5	Code Comments	Important steps are explained with comments	Yes	Explained logic
6	Modular Design	Code divided into reusable functions	Yes	Functions reused
7	No Redundant Code	No duplicate or unnecessary code present	Yes	Optimized
8	Error Handling	Basic input validation implemented	Partial	Basic handling added

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where used	Reusability Level (High / Medium / Low)
1	Data Preprocessing Function	Python	Model Training	High
2	Prediction Function	Python	Web App	High
3	Visualization Module	Matplotlib/Seaborn	Analysis	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Properly divided into sections
Readability	4	Easy to read and understand
Reusability	4	Functions reused in multiple places
Documentation / Comments	3	Some comments added, can improve
Overall Score	4	Good overall implementation